Long-term Time Series Prediction of Deformation in The Area of Pylons By Combining InSAR and Transformer-based

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***Abstract*—Settlement monitoring of high-voltage line pylons (HLPs) is an important part of power line inspection. In geologically active areas, or due to some environmental factors, it can substantially impact the stability of HLP bases, which can lead to collapse accidents, so it is essential to monitor and predict the areas around HLPs. In this paper, we use 74-view Sentinel-1A satellite ascending orbit data, and combine the persistent scatterer interferometric synthetic aperture radar (PS-InSAR) with a long-term sequence prediction model (Autoformer) to perform surface deformation of a high-voltage line in Inner Mongolia, China, from January 2019 to January 2022 monitoring. The environmental factors were considered to predict the surface deformation in the key study area from February 2022 to April 2022 and provide early warning. By comparing with Transformer, Informer, and Reformer, the results show that the proposed combined PS-InSAR and Autoformer method is far superior to its counterparts with a mean squared error of 0.105 . This paper provides a new, accurate, and automated method for HLP subsidence monitoring and early warning and establishes a system for HLPs surface deformation prediction, which highly improves the efficiency of HLP monitoring and early warning.**

Keywords—Transformer, Interdisciplinary method, Surface deformation, InSAR, Long-time series prediction

I. INTRODUCTION

Surface deformation is a long-term, progressive geohazard [1]. Among them, high-voltage line pylon(HLP) regional surface deformation caused by some continuous natural changes, such as environmental factors including temperature, wind speed[2], soil moisture, rainfall[3], etc., or some human-induced factors will cause HLP damage such as increasing the tension of cables and the possibility of the breaking of the lines or tilting of pylons[4]will cause severe grid accidents, endanger the power transmission system and pose a serious threat to human life and property safety. HLP is the basic production guarantee of the power system, maintaining the normal production of human life and the safe operation of the power grid construction is essential. Li et al.[5] used ALOS-2 PALSAR radar satellite data to monitor the surface deformation near the transmission channel of the Jinsu line in Sichuan using the differential interference method and found the existence of obvious landslide hazards in the vicinity, which effectively reduced the economic loss. Therefore, it is essential to monitor, analyze and provide early warning for the spatial and temporal changes in the pylon base area of HLP.

The time series prediction problem is to predict the future length O series based on the past length-I series, denoted as Input-I-predict-O. Surface deformation prediction is an application of time series prediction. Most of the existing surface deformation prediction studies are based on traditional neural network models and image data. Among them, Ma et al.[6]used COSMO-SkyMed images and the PS-InSAR technique to conduct the Hong Kong International Airport monitored the Hong Kong International Airport and constructed a CNN model to predict the linear deformation of the airport land and the nonlinear expansion of buildings. Reddy et al.[7]used NDVI time series data and the LSTM model to predict the vegetation cover dynamics. All of the above studies obtained good prediction accuracy compared to InSAR. The above studies further demonstrate the usefulness and effectiveness of neural networks in time series prediction. However, CNNs are not fully applicable to learning time series and require various auxiliary processing. LSTM has some disadvantages in parallel processing, and the structure of LSTM determines that they lead to information loss during sequential computation and do not apply to longer time series data.[8]

Surface deformation is continuous and slow, and long-term surface deformation prediction has become challenging. The Transformer[9]model proposed by Vaswani largely addresses these issues by implementing a self-attentive mechanism that avoids recursion and allows parallel computation, thereby reducing training time and performance degradation due to long-term dependencies. Many Transformer models have been successfully applied to time series prediction tasks[10]. For example, the work of Li et al.[11]and Zhou et al.[12] applied transformer to time series prediction of open-source datasets in multiple industries. All of the above studies have demonstrated its effectiveness in time series and long-term time series forecasting tasks.Due to the excellent performance of Transformer-based machine learning models in long-term time series prediction tasks and the lack of current studies applying them to HLP regional InSAR surface deformation long-term time series prediction.

Therefore, this study demonstrates the workflow of surface deformation monitoring, modeling, and prediction in a HLP regional surface in the Inner Mongolia Autonomous Region, China, by combining InSAR processing, environmental factor data, and Autoformer models. By obtaining 2m air temperature, 2m specific humidity, 10m wind speed, hourly precipitation, soil temperature, and humidity data from national meteorological automatic stations with regular observations and combining InSAR surface deformation time series data as the input of the prediction model, a long-term series prediction model (Autoformer) based on the depth decomposition architecture and autocorrelation mechanism is used to simulate and predict the deformation. Also, the prediction results of Autoformer[13]were compared with Transformer[9],

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Informer[12], and Reformer[14]. The model performance of Autoformer is significantly better than other models, proving the effectiveness and robustness of the model in the aspect of predicting of deformation.

II. Study area and data set

The study area of this paper is located in the Inner Mongolia Autonomous Region of China, which is situated in the northern part of China. The soil loss caused by windstorms and dust and the drastic changes in the ground surface caused by heavy rainfall, flooding, and mudslides can have a substantial impact on the stability of the base of HLP, which can lead to severe pylon tilt or pylon collapse accidents, causing significant economic loss[15]. Therefore, it is imperative to carry out accurate monitoring and prediction of regional surface deformation in HLP in the Inner Mongolia Autonomous Region. The high-voltage transmission line studied in this paper is very significant to local people's livelihood and production. The total length of the transmission line is 68km, including 123 pylons. This paper selects the 500m buffer zone on both sides of the line as the study area.

Sentinel-1A orbit-raising images provided by the European Space Agency (ESA). The period is January 2019 to April 2022. The polarization Sentinel-1A satellite image data were used in this study. The data are 94-view C-band n mode is VV, and the imaging mode is IW. To improve the image orbit accuracy, the POD precision orbit ephemeris data are introduced in this study, and the external DEM uses 30m grid spacing SRTM data. All of the above data have good data quality.

III. Experimental Methods

In this subsection, the PS-InSAR technique is applied to process the Sentinel-1A satellite image

data of a HLP in the Inner Mongolia Autonomous Region, China, and the InSAR surface deformation time series are determined. Data pre-processing was performed on the environmental factor data set. Based on this, Autoformer, Transformer, Informer, and Reformer prediction models were established and compared, and the Autoformer was applied to the surface deformation prediction in the study area. The method flow is shown in Figure 1.

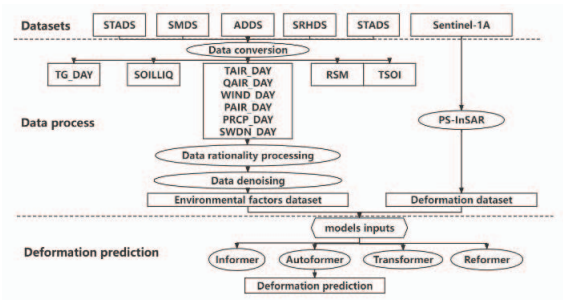

Figure 1. Experimental method.

A. PS-InSAR

The PS-InSAR method was proposed by Ferretti et al.[16] and has been widely used for surface deformation monitoring with high accuracy. The PS-InSAR method, by selecting point targets with stable scattering characteristics at the ground surface, such as rocks, manufactured structures, etc. The interference of non-stable point targets on the inversion results is avoided. Based on the differential phase information of these discrete points, a phase function model is established, and a rational model algorithm is adopted to weaken the effect of out-of-correlation and separate the delayed atmospheric phase to obtain time-series surface deformation data. It is based on long time series, effectively weakening the effects of out-of-correlation and atmospheric delay, significantly reducing the effects of temporal and spatial out-of-correlation, and achieving the leap of monitoring accuracy from centimeter level to millimeter level, which can perform accurate and reliable surface information inversion.

B. Autoformer

Autoformer[13] is a deep learning network model based on deep decomposition architecture and autocorrelation mechanism. Aiming at the problems of complicated processing of complex temporal patterns and high operational efficiency in long-term series prediction, it substantially improves long-term series prediction efficiency through progressive decomposition and sequence-level connectivity. Autoformer has shown excellent long-term prediction results in five mainstream fields: energy, transportation, economy, meteorology, and disease. Since surface deformation is continuous and slow, long-term prediction of surface deformation is necessary. The long-term series surface deformation prediction is becoming a challenge. The superior results of Autoformer for long-term series prediction make it suitable for the long-term series prediction problem of surface deformation.

The Autoformer network structure is shown in Figure 2 and contains internal serial decomposition blocks, an autocorrelation mechanism, and corresponding decoders and encoders. Where the encoder eliminates the long-term trend cycle part through the serial decomposition block (blue block) and focuses on seasonal pattern modeling. The decoder progressively accumulates the trend component extracted from the hidden variables. The encoder-decoder autocorrelation uses past seasonal information in the encoder (green in the middle of the decoder).

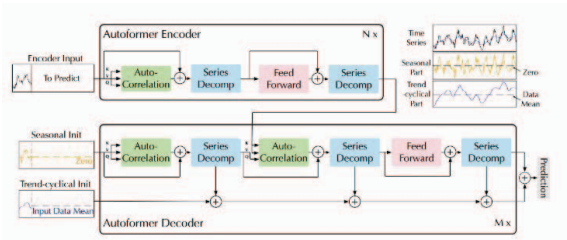

Figure 2. Autoformer network structure.

Time series decomposition refers to the decomposition of a time series into several components, each of which represents a class of potential temporal patterns, such as seasonal, trend-cyclical terms. Due to the unknowability of the future in prediction problems, it is common to first decompose the past sequence and then predict it separately. However, this causes the prediction results to be limited by the decomposition effect and ignores the interactions between the future components. As shown in Figure 2, autoformer proposes a deep decomposition architecture that embeds sequence decomposition into the encoder-decoder as an internal unit of Autoformer.

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IV. RESULTS

A. Spatial and temporal distribution of surface deformation in the study area

Based on 74 ascending orbit Sentinel-1A images, 9894 PS points were obtained from the InSAR time series surface deformation results for January 2019 to January 2022 using PS-InSAR analysis using SARPROZ. A total of 300 PS points in the study area were selected from the InSAR time-series surface deformation results, as shown in Figure 3.

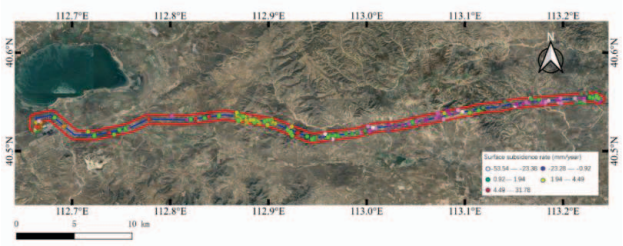

Figure 3. Results of surface deformation in the study area.

The spatial distribution of surface deformation in the study area is uneven, with a trend of gradual subsidence from the west to the east. In this paper, we analyze the ten PS points in the study area with the most significant surface deformation rates. The time distribution of accumulated subsidence at study sites A, B, C, D, and E and the time distribution of accumulated uplift at study sites 1, 2, 3, 4, and 5 are shown in Figure 4. In this study, the above ten study points are classified as high-risk deformation sites, which should be investigated in the field and predicted timely.

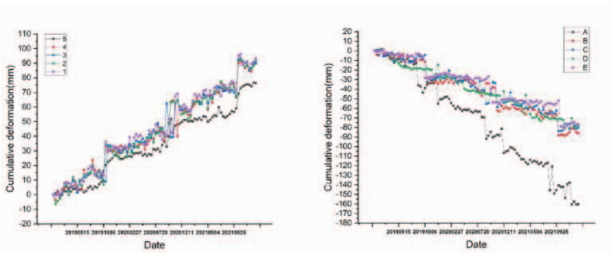

Figure 4. Time distribution of cumulative shape variables at study sites.

B. Prediction of surface deformation in the study area

1) Simulation results. To predict the upcoming surface deformation more accurately, avoid losses to HLPs caused by surface deformation. In this study, environmental factor parameters and InSAR surface deformation time series were used as model inputs, Autoformer, Transformer, Informer, and Reformer, were modeled and trained with different scaled steps. The training set accounts for 80% of the dataset and the test set accounts for 20% of the dataset. To determine the performance under different future applications, a series of model comparison experiments with input length to prediction length ratios of 1:2, 1:3, 1:4, and 1:5 were conducted, respectively, and the experimental step settings in this paper are consistent with the definition of long-term prediction. The performance of each model is shown in Figure 5.

The MSE and MAE values of Autoformer are significantly better than those of other models in the InSAR surface deformation time-series dataset, and the performance of Autoformer changes fairly steadily with increasing prediction level, which means that Autoformer has better long-term prediction robustness. This demonstrates the effectiveness and superiority of Autoformer in predicting surface deformation.

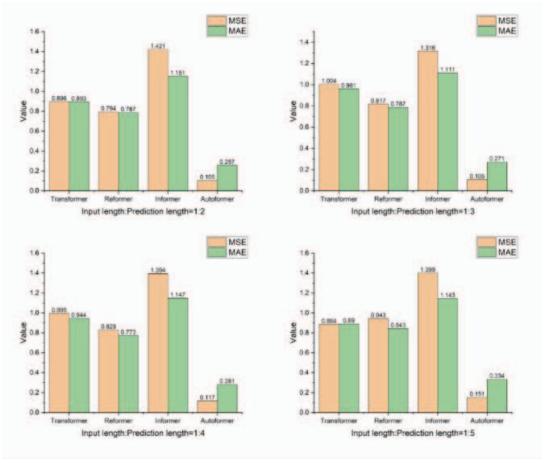

Figure 5. Model performance comparison.

2) Prediction results. A total of 760 consecutive surface deformation monitoring values from January 1, 2019 to January 30, 2022 were used to train the Autoformer model in this study. After verifying the reliability of the Autoformer model, the surface deformation time series prediction was performed for study point B from February 1, 2022 to May 1, 2022 with a prediction length of 90 for the input step of 30 predictions from January 2, 2022 to January 31, 2022. the distribution of the predicted time series is shown in Figure 6. The predicted value of this area reaches the cumulative settlement of 88.43 mm on April 29, 2022, which is a certain threat to HLP and may lead to HLP tilting or collapsing accidents, and should be investigated in the field to eliminate hidden risks.

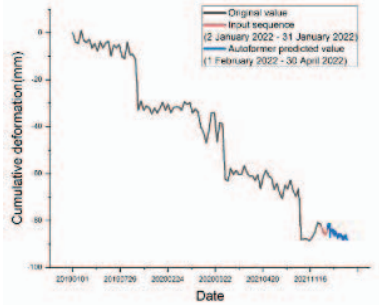

Figure 6. Predicted surface deformation in the area of study point B.

V. CONCLUSION

In this study, time-series data of InSAR surface deformation of HLPs in the Inner Mongolia Autonomous Region from January 2019 to January 2022 were obtained using the PS-InSAR analysis technique based on Sentinel-1A images of ascending orbit. Combined with the environmental factors in the region, the Autoformer prediction model was established to predict the time series of surface deformation for the study area and compared with the Transformer, Informer, and Reformer prediction models. The model performance of Autoformer is 0.105 for MSE and 0.257 for MAE, which is significantly better than other models, proving the effectiveness and robustness of the model.

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Autoformer combined with InSAR is a very accurate and effective method applied to time series prediction of surface deformation and used for the regional monitoring and warning system of HLPs. In the future, more factors can be incorporated into the model to more accurately simulate and predict the surface deformation to improve the accuracy and validity of the model prediction. The monitoring and prediction methods of HLPs in this paper can effectively and timely detect surface deformation in HLP areas, greatly reduce the damage to HLPs caused by surface deformation, improve the inspection efficiency of HLPs, and maintain the safe operation of the grid system.

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